



FGITrans: Cross-City Transformer for Fine-grained Urban Flow Inference

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Abstract

Inferring the fine-grained urban flows based on the coarse-grained flow observations is practically important to many smart city-related applications. Adequate data is usually a prerequisite for existing machine learning methods, especially most deep learning models. However, many cities still suffer from the data scarcity issue due to the unbalanced city development levels. To mitigate this issue, we propose a novel cross-city fine-grained urban flow inference model named FGITrans, which aims to effectively transfer the knowledge from the data-rich cities to the data-scarce cities. Specifically, we design a weight-sharing triple-branch transformer framework which adopts self-attention and cross-attention for source/target city feature learning and domain alignment, respectively. Then, we propose a novel spatio-temporal adaptive embedding (STAE) layer for our transformer framework, and introduce a cross-city knowledge distillation (CKD) loss to narrow the cross-city disparities. The CKD loss explicitly enforces the framework to learn the discriminative domain-specific and domain-invariant representations simultaneously. Extensive experiments conducted on four large real-world datasets validate the effectiveness of FGITrans compared with the state-of-the-art baselines.

CCS Concepts

• **Information systems** → **Location based services.**

Keywords

Spatial-temporal data mining, Urban flow inference, Transfer learning

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1 Introduction

Fine-grained urban flow inference is practically important for smart city construction and intelligent traffic systems (ITS). Accurately depicting the fine-grained patterns of urban flows serves as the foundation of numerous smart city services, such as trip planning [1], resource management [2], and traffic prediction [3, 4]. To obtain the fine-grained urban flow observations, a large number of sensors are required to be deployed at different locations of a city. However, in reality, the number of deployed urban flow sensors can be quite limited due to the high expenditure in terms of daily operation and maintenance. Therefore, how to effectively infer the fine-grained urban flows based on the sparse and coarse-grained observations has become an important research issue, attracting rising research attention [5, 6].

Recently, inspired by the success of deep neural networks, deep learning based methods are emerging and achieve remarkable performance gains in fine-grained urban flow inference [7, 8]. While deep learning models have significantly improved the performance, they expose unignorable problems. Specifically, the superior performance of deep learning models highly relies on large-scale training data such as the massive sensors records and the auxiliary semantic information, which are usually inaccessible in real application scenarios. In reality, the data-scarce issue is common due to various reasons such as sparse sensors deployment, data privacy concerns, and the low city development level [9]. For instance, the developed cities like Beijing and New York have released large taxi datasets for a couple of years, while the traffic data collected from the newly built cities such as Xiongan New Area only covers a few months. As a result, existing approaches trained on data-rich cities (e.g., Beijing and New York) are hard to accommodate those cities with sparse data [10]. Therefore, how to enhance the performance of fine-grained urban flow inference for the cities with insufficient



Figure 1: Illustration of inferring the fine-grained urban flows in a data-scarce city with the help of the knowledge transferred from a data-rich city.

data by incorporating the knowledge transferred from the cities with sufficient data is of pressing importance.

To address this issue, this paper proposes a novel transfer learning framework which targets at inferring the fine-grained urban traffic flows in data-scarce cities via the knowledge transferred from data-rich cities. However, it is a formidable task to design such a domain adaptation approach due to the following challenges.

- **Insufficient integration of domain-invariant and domain-specific knowledge.** Existing approaches typically separately design sub-modules to capture domain-invariant and domain-specific features across different cities [9, 11, 12]. Consequently, when utilizing both features for inference in the target city, insufficient feature integration often results in suboptimal inference outcomes. Therefore, the design of a framework which is capable of simultaneously capturing both domain-invariant and domain-specific features is of paramount importance.
- **Significant discrepancy between source and target cities.** The significant difference in terms of data distribution between source and target cities may cause unstable performance and increase the risk of negative transfer [13]. Thus, how to design a method that is capable of mitigating the discrepancy between the data distributions of the source and target cities is also challenging.

With the success of transformer architecture in natural language processing (NLP) [14–17] and computer vision tasks [18–21], cross-attention in transformer has been proved to work quite well at aligning different distributions, even from different data modalities [22, 23]. Inspired by the success of transformer, this paper proposes a novel transformer-based fine-grained urban flow inference model FGITrans, which is characterized by effective cross-city knowledge transfer. Specifically, we first introduce a triple-branch transformer with shared weights. The first two self-attention branches aim to model the source and the target city data respectively, and the third cross-attention branch aligns the two domains by the well-designed knowledge integration module. Then, we propose a cross-city spatio-temporal transformer backbone with spatio-temporal adaptive embedding (STAE) layer for obtaining the spatial and temporal features of each branch. Finally, we devise a cross-city knowledge distillation (CKD) loss to facilitate the alignment of the temporal and spatial features extracted from different cities. Such design explicitly enforces the framework to learn the discriminative

domain-specific and domain-invariant representations simultaneously.

In summary, this paper makes the following contributions.

- We for the first time study the fine-grained urban flow inference problem in a data-scarce city by transferring knowledge from a data-rich city, and design a transformer based framework named FGITrans to effectively address it.
- FGITrans is designed as a weight-sharing triple-branch transformer, where the backbone of each branch is a cross-city spatio-temporal transformer with spatio-temporal adaptive embedding (STAE) layer.
- We introduce the cross-city knowledge distillation (CKD) loss to facilitate the alignment of features between source and target cities.
- Extensive experiments are conducted on four large real-world datasets. Experimental results demonstrate the superiority of FGITrans by comparison with existing state-of-the-art approaches.

The remainder of the paper is organized as follows. We will first briefly review related work in Section 2. Notations and problem definition will be introduced in Section 3. FGITrans framework will be introduced in Section 4, followed by experimental results in Section 5. Finally, we will conclude the paper in Section 6.

2 Related Work

2.1 Fine-grained Urban Flow Inference

Motivated by the super-resolution methods in computer vision (CV), methods like SRCNN [24], VDSR [25], and SRResNet [26] have been proposed to address the spatial-temporal data super-resolution task and have achieved superior performance than traditional methods [27]. However, directly borrowing the super-resolution methods from the CV area will fall short in ignoring the complex spatial-temporal correlations of the traffic data and some external factors such as weather, temperature, and holidays. To address this limitation, methods like UrbanFM [1], UrbanPy [7], and DeepLGR [28] have been proposed, which incorporate the specific characteristics of urban traffic flow data into the model design and achieve significant performance improvement. To address the issue that the coarse-grained urban flows collected in real-world scenarios can be sparse and incomplete, MT-CSR [8] establishes a joint network for both urban flow data imputation and super-resolution. Considering that the coarse-grained urban flows collected in real-world scenarios are not always reliable, Zheng et al. [29] proposes a Diffusion-based robust fine-grained urban flow inference model named DiffUFlow that can conduct data denoising and super-resolution simultaneously.

2.2 Domain Adaptation

Most domain adaptation methods tackle the domain shift issue by aligning the feature distributions. Early domain adaptation methods such as [30–35] adopt moment matching to align feature distributions. In recent years, plenty of works have emerged that achieve alignment by adversarial training. DANN [36] formulates domain adaptation as an adversarial two-player game. The adversarial training of CDAN [37] is conditioned on several sources of information. DIRT-T [38] performs domain adversarial training with an added

term that penalizes violations of the cluster assumption. Additionally, [39–41] adopts prediction diversity between multiple learnable classifiers to achieve local or category-level feature alignment between source and target domains. AFN [42] shows that the erratic discrimination of target features stems from much smaller norms than those found in the source features. SRDC [43] is proposed to directly uncover the intrinsic target discrimination via discriminative clustering to achieve adaptation. More related, [44] resorts to K-means clustering for open-set adaptation while considering the global structure. Different from the above methods, our method focuses on the nearest neighbors (local structure) for source-free adaptation.

2.3 Transfer Learning for Traffic Forecasting

Data insufficiency is a common problem in real-world traffic forecasting tasks. For example, for newly deployed traffic sensors, it usually needs a long time to accumulate the large-scale traffic data, during which many smart city services may be compromised. To address the problem, transfer learning methods have been proposed for traffic forecasting. For example, RegionTrans [9], MetaST [13], and CrossTRES [11] are proposed to transfer knowledge for grid-based urban flow data, and are thus not optimal for graph-based traffic data. STGFSL [45], DASTNet [12] and TransGTR [46] are designed for graph-based traffic data. Motivated by the success of transfer learning for traffic forecasting, this paper for the first time designs a novel domain adaptation model with knowledge transfer for fine-grained urban flow inference across different cities.

3 Notations and Problem Definition

We will first give some definitions to help state the studied problem, and then present a formal problem definition.

DEFINITION 1. Cell region. We divide a city into a grid map consisting of $I \times J$ cell regions based on latitude and longitude. We denote all the cell regions as $R = \{r_{1,1}, \dots, r_{i,j}, \dots, r_{I,J}\}$, where $r_{i,j}$ is the i -th row and j -th column cell region of the grid map.

DEFINITION 2. Urban flow map. Let $\mathcal{T}$ represent a collection of urban flow trajectories. When considering a specific cell region $r_{i,j}$, the associated inflow and outflow map of urban flows during the time slot t can be defined as follows,

$$X_{in,i,j}^t = \sum_{f^t \in \mathcal{T}} \{f^{t-1} \notin r_{i,j} \cap f^t \in r_{i,j}\}$$

$$X_{out,i,j}^t = \sum_{f^t \in \mathcal{T}} \{f^t \in r_{i,j} \cap f^{t+1} \notin r_{i,j}\}$$

where $f \in \mathcal{T}$ represents a trajectory within the collection. $f^{t-1} \notin r_{i,j}$ indicates that the trajectory f at time $t-1$ does not fall into the region $r_{i,j}$. Conversely, $f^t \in r_{i,j}$ denotes that the trajectory f at time t falls into the region $r_{i,j}$. $\cap$ represents the intersection operator. To represent the inflows and outflows of all regions at time t , we introduce the urban flow tensor $x^t \in \mathbb{R}^{2 \times I \times J}$.

DEFINITION 3. Urban semantic variables. Urban semantic variables include external factors and land features. External factors contains temperature, windspeed, holiday, et al. We represent these external factors in a time slot t as a vector $\mathcal{V}_{sem}^t \in \mathbb{R}^l$, where l is the feature length. We collect POI category distribution and the structural

attributes of road networks in each cell region as its land features, represented as a matrix $\mathcal{P} \in \mathbb{R}^{K \times I \times J}$, where K is the feature category.

DEFINITION 4. Coarse- and fine-grained urban flow map. A coarse-grained urban flow map represents the observed urban flows derived from the flow sensors. It is generated by integrating neighboring grids within an $N \times N$ range from a fine-grained urban flow map, where N is the upscaling factor. We denote the coarse- and fine-grained urban flow maps at time t as $x_{cg}^t \in \mathbb{R}^{2 \times I \times J}$ and $x_{fg}^t \in \mathbb{R}^{2 \times NI \times NJ}$, respectively. We define the spatio-temporal urban flow time series as follow:

$$\mathcal{X}_{\mathcal{D}} = \{\mathcal{X}_{t,\mathcal{D}} | t \in \mathcal{T}_{\mathcal{D}}\} \in \mathbb{R}^{T_{\mathcal{D}} \times 2 \times (N)I \times (N)J}$$

In this paper, the target city $\mathcal{D}$ suffers from the sensor data scarcity issue, while the source city $\mathcal{D}'$ has rich sensor data, i.e., $|\mathcal{T}_{\mathcal{D}}| \ll |\mathcal{T}_{\mathcal{D}'}|$. Based on the above definitions, we formulate the problem as follows.

Problem Statement. Given an upscaling factor N , a small amount of data $\mathcal{X}_{\mathcal{D}}$ in target city $\mathcal{D}$ and rich data $\mathcal{X}_{\mathcal{D}'}$ in source city $\mathcal{D}'$, coarse-grained urban flow spatial maps $x_{cg}^t \in \mathbb{R}^{2 \times I \times J}$ and urban semantic variables $\mathcal{V}_{sem}$ and $\mathcal{P}$ from two cities, we aim to infer the fine-grained urban flow map $x_{fg}^t \in \mathbb{R}^{2 \times NI \times NJ}$ in the target city.

4 Methodology

4.1 Framework Overview

In this section, we will elaborate the novel cross-city transfer learning model FGITrans for the fine-grained urban flow inference, and the model framework is shown in Figure 2. Specifically, we first propose a triple-branch weight-sharing transformer framework, two branches of which are for source and target city data respectively and the third one is the feature alignment branch, whose inputs are from source-target city pairs. The self-attention is applied in the source/target transformer branches and the cross-attention is involved in the feature alignment branch to conduct domain alignment. Such a design explicitly enforces the framework to learn discriminative domain-specific and domain-invariant representations simultaneously. We will elaborate this part in section 4.2. Then in each branch, we design a cross-city spatio-temporal transformer with spatio-temporal adaptive embedding (STAE) layer as its backbone for obtaining the spatial and temporal features. We will explicate it in detail in section 4.3. Finally, we introduce a cross-city knowledge distillation (CKD) loss between the source branch and the target branch to enforce constraint over intermediate temporal and spatial features extracted across the two cities, which will be described in section 4.4.

4.2 Weight-Sharing Triple-Branch Transformer

FGITrans contains three weight-sharing transformers with corresponding data flows and constraints. The three branches are source city branch, target city branch, and alignment branch. As shown in Figure 2, the source and target spatio-temporal urban flow maps in the input pair are sent to source city branch and target city branch, respectively. In the two branches, the self-attention module is used to learn the domain-specific representations. The Pseudo-Huber

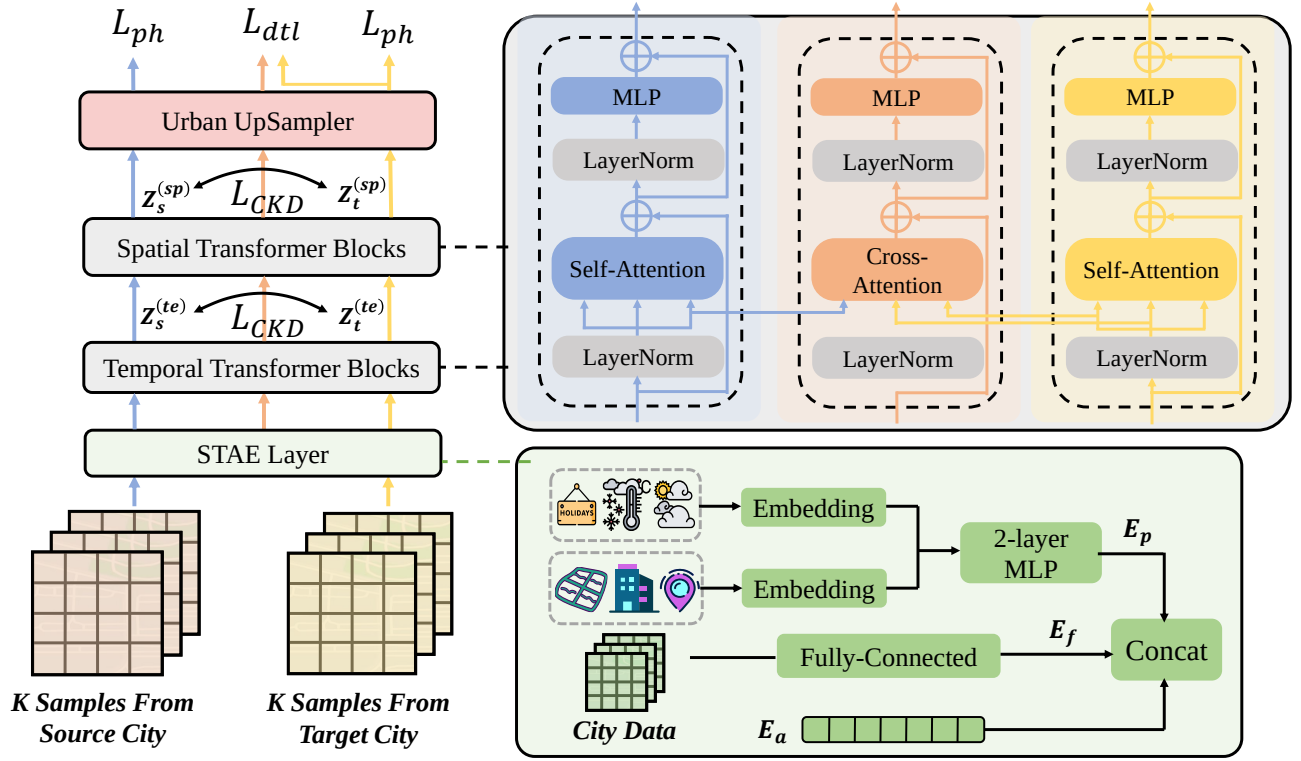


Figure 2: The proposed FGITrans framework consists of three weight-sharing spatio-temporal transformers with the urban flow maps from the source and target cities. Pseudo-Huber loss (L_{ph}) is adopted to the source and target branches. The distillation loss (L_{dtl}) is applied between the alignment branch and the target branch. Cross-city knowledge distillation loss (L_{CKD}) is utilized to align intermediate spatio-temporal representations between the source and target branches.

loss defined as follows is used to train the inference accuracy,

$$L_{ph} = \sum_{i=0}^{I \times J} \delta^2 \left(\sqrt{1 + \left(\frac{y^i - \hat{y}^i}{\delta} \right)^2} - 1 \right), \quad (1)$$

where y^i is the inference result of each branch and $\hat{y}^i$ is the ground truth. δ is an adjustable hyperparameter to balance the accuracy and robustness of the model.

The cross-attention module is imported in the alignment branch. The inputs of the alignment branch are from the other two branches. The query of the cross-attention module comes from the query of the source branch, while the keys and values are from the target branch. Then the cross-attention module outputs the aligned features which are added with the output of the next layer.

We use the output of the alignment branch to guide the training of the target branch. Specifically, the alignment branch and the target branch are denoted as the teacher and the student, respectively. We consider the inference result of the upsampler in alignment branch as a soft label that can be used to further supervise the target branch through L_{dtl} :

$$L_{dtl} = \sum_{i=0}^{I \times J} \delta^2 \left(\sqrt{1 + \left(\frac{q^i - p^i}{\delta} \right)^2} - 1 \right), \quad (2)$$

where q^i and p^i are the inference results from the alignment branch (teacher) and the target branch (student) respectively.

During inference, only the target branch is used. The input is an image from target city, and only the target data flow is triggered, i.e. the yellow lines in Figure. 2. Its output of the upsampler is utilized as the final inference labels.

4.3 Cross-City Spatio-Temporal Transformer

In each branch mentioned above, we propose a cross-city spatio-temporal transformer backbone which enjoys a compact architecture and compelling efficacy in modeling the spatio-temporal dependency of multiple attributes.

We add an embedding layer for the input, providing multiple types of embeddings for the model backbone. To further enhance the representation effectiveness, we propose a novel spatio-temporal adaptive embedding (STAE) layer and apply it on the cross-city spatio-temporal transformer.

In the STAE layer, to preserve the information from the raw data, we utilize a fully connected layer to obtain the feature embedding $E_f \in \mathbb{R}^{T \times H \times W \times d_f}$, where d_f is the dimension of the feature embedding. External factor features contain weather, windspeed, temperature, et al. We denote the learnable day-of-week embedding as $T_w \in \mathbb{R}^{N_w \times d_f}$, and the time stamps-of-day embedding as

$T_d \in \mathbb{R}^{N_d \times d_f}$, where $N_w = 7$ denotes the number of days in a week and $N_d = 288$ is the number of timestamps in each day. We use them as indices to extract the corresponding day-of-week embedding $E_w^t \in \mathbb{R}^{T \times d_f}$ and timestamp-of-day embedding $E_d^t \in \mathbb{R}^{T \times d_f}$ from the embedding dicts. By concatenating and broadcasting them, we get the periodicity embedding $E_p \in \mathbb{R}^{T \times H \times W \times 2d_f}$ for the urban flow time series. For spatial relation modeling, we design a spatio-temporal adaptive embedding $E_a \in \mathbb{R}^{T \times H \times W \times d_a}$ to capture the intricate spatio-temporal relation in a uniform way instead of adopting a pre-defined or dynamic adjacency matrix. In particular, E_a is shared across different traffic time series. By concatenating the embeddings above, we obtain the hidden spatio-temporal representation $Z_{in} = E_f || E_p || E_a$.

We apply transformer along temporal and spatial dimensions to capture the intricate traffic flow relations. Given the hidden spatio-temporal representation, the transformer input is $Z_{in} \in \mathbb{R}^{T \times H \times W \times d_h}$ with T frames and $H \times W$ spatial grids. We obtain the query, key and value matrices through temporal transformer layers as follows,

$$Q^{(te)} = Z_{in} W_Q^{(te)}, K^{(te)} = Z_{in} W_K^{(te)}, V^{(te)} = Z_{in} W_V^{(te)}, \quad (3)$$

where $W_Q^{(te)}, W_K^{(te)}, W_V^{(te)}$ are learnable parameters. We obtain the output of temporal transformer $Z^{(te)}$ as follows,

$$Z^{(te)} = \text{Softmax} \left(\frac{Q^{(te)} K^{(te)T}}{\sqrt{d_h}} \right) V^{(te)}. \quad (4)$$

Similarly, the spatial transformer layer performs as follows,

$$Z^{(sp)} = \text{Attn}_{self}(Z^{(te)}). \quad (5)$$

The cross-attention module is derived from the self-attention module. The difference is that the input of cross-attention is from two branches.

In order to bring the spatio-temporal features extracted from the target branch closer to those of the source branch, we introduce L_{CKD} constraint on $Z^{(te)}$ and $Z^{(sp)}$, which will be described in section 4.4.

4.4 Cross-City Knowledge Distillation

Despite the use of L_{ph} and L_{dtl} to constrain the fine-grained inference results at the output level, it is also essential to align the intermediate representations of the source branch and the target branch. Thus we aim to achieve a better feature extraction by constraining the intermediate representations $Z^{(te)}$ and $Z^{(sp)}$, respectively. Specifically, we implement a cross-city knowledge distillation (CKD) loss between the two branches to enforce alignment over temporal and spatial representations extracted from the source and target cities. The goal of the CKD loss is to shrink the cross-city discrepancy and improve the generalization ability of the network so that the spatio-temporal features from the source branch are similar with those from the target branch. The CKD loss contains three parts, the city-aware distillation (CD) loss, the pair-wise constraint (PC) loss, and the maximum mean discrepancy (MMD) loss.

City-aware Distillation Loss: The source branch and the target branch share the same model architecture, but they focus on feature extraction in different cities. These two branches can be considered as the teacher and student networks in the context of knowledge

distillation learning. Therefore, we propose a city-aware distillation (CD) loss to narrow the cross-city gap by transferring city-aware knowledge from the source city to the target city. The CD loss ensures the two branches produce similar features with consistent distribution across different cities. Particularly, the CD loss make the student network mimic the teacher network by using the KL divergence loss, and the loss function $\mathcal{L}_{CD}$ can be written as follows,

$$\mathcal{L}_{CD} = \frac{1}{N} \sum_{i=1}^N T^2 \mathcal{L}_{KL}(\sigma(\frac{Z_t}{T}), \sigma(\frac{Z_s}{T})), \quad (6)$$

where Z_t and Z_s represent the intermediate representations from the target and source cities respectively, encompassing both $Z^{(te)}$ and $Z^{(sp)}$. $\sigma(\cdot)$ refers to the softmax function, and T is the temperature hyperparameter to smooth the distillation loss. The KL divergence loss is written as follows,

$$\mathcal{L}_{KL}(P(Z_t), P(Z_s)) = \sum P(Z_t) \log \left(\frac{P(Z_t)}{P(Z_s)} \right), \quad (7)$$

where $P(Z_t) = \sigma(\frac{Z_t}{T})$ and $P(Z_s) = \sigma(\frac{Z_s}{T})$.

Pair-Wise Constraint: The CD loss employs an indirect solution to make the target branch predict features with similar distribution as the source branch. Despite the CD loss, we need to apply numerical restraint to directly transfer knowledge in the feature space, making the features of the two branches more similar. Specially, we propose a pair-wise constraint (PC) as follows to make the target branch imitate the source branch directly by using Pseudo-Huber loss,

$$\mathcal{L}_{PC} = \sum_{i=0}^{I \times J} \delta^2 \left(\sqrt{1 + \left(\frac{Z_t^i - Z_s^i}{\delta} \right)^2} - 1 \right). \quad (8)$$

Maximum Mean Discrepancy Loss: We employ the Maximum Mean Discrepancy (MMD) loss to tackle with the cross-resolution problem because MMD is effective for shrinking the discrepancies in changing cities. Given the corresponding source branch features and the target branch features, the MMD loss can be formulated as:

$$\mathcal{L}_{MMD} = \left\| \frac{1}{N} \sum_{i=1}^N \phi(Z_t) - \frac{1}{N} \sum_{j=1}^N \phi(Z_s) \right\|_2^2, \quad (9)$$

where $\phi(\cdot)$ denotes the projection function. Derived from [47], the MMD loss can be written as follows,

$$\begin{aligned} \mathcal{L}_{MMD} = & \left\| \frac{1}{N^2} \sum_{i,j=1}^N k(Z_t^i, Z_t^j) - \frac{2}{N^2} \sum_{i,j=1}^N k(Z_t^i, Z_s^j) \right. \\ & \left. + \frac{1}{N^2} \sum_{i,j=1}^N k(Z_s^i, Z_s^j) \right\|_2^2, \end{aligned} \quad (10)$$

where $k(\cdot)$ denotes the kernel function. Particularly, we adopt the Gaussian kernel to measure the distance between the source city and target city features, which can be formulated as:

$$k(Z_t^i, Z_s^j) = e^{-\|Z_t^i - Z_s^j\|_2^2 / 2\sigma^2}. \quad (11)$$

Finally, the overall cross-city knowledge distillation loss can be formulated as follows,

$$\mathcal{L}_{CKD} = \mathcal{L}_{CD} + \alpha \mathcal{L}_{PC} + \beta \mathcal{L}_{MMD}, \quad (12)$$

where α and β are the trade-off hyperparameters.

Table 1: Dataset description

Dataset	BJTaxi	NYCTaxi	NYCBike	CDdidi
Latitude	/	(40.71, 40.765)	(40.71, 40.765)	(30.655, 30.727)
Longitude	/	(-74.01, -73.972)	(-74.01, -73.972)	(104.043, 104.129)
Time span	3/1/2015-6/30/2015	1/1/2015-12/31/2015	1/1/2015-12/31/2015	11/1/2016-11/31/2016
Time interval	30 minutes	1 hour	1 hour	1 hour
Size	5760	8760	8760	720
Upscaling	$16 \times 16 \rightarrow 32 \times 32$			

Table 2: The performance comparison of different methods over the four datasets. The first type of baselines use 20% of the target city data for training. The second type of baselines are first trained on the entire source city data and then fine-tuned with 20% of the target city data

		Target data only			Source & Target data				Improve
		UrbanPy	MT-CSR	DiffUFlow	UrbanPy(FT)	MT-CSR(FT)	DiffUFlow(FT)	FGITrans	
BJTaxi→NYCTaxi	MAE	0.47	1.62	1.03	<u>0.43</u>	1.58	0.97	0.41	4.65%
	RMSE	0.85	2.10	1.50	<u>0.78</u>	1.95	1.39	0.51	34.61%
	MAPE	0.31	0.72	0.61	<u>0.29</u>	0.70	0.58	0.27	6.90%
NYCTaxi→NYCBike	MAE	0.36	2.01	1.51	<u>0.25</u>	1.33	1.16	0.23	8.00%
	RMSE	2.53	3.47	3.28	<u>1.65</u>	2.30	2.11	0.92	44.24%
	MAPE	0.07	0.89	0.87	<u>0.06</u>	0.85	0.84	0.06	0.00%
NYCBike→CDdidi	MAE	1.78	2.26	1.69	<u>1.39</u>	2.15	1.55	1.15	17.27%
	RMSE	3.63	3.92	2.93	<u>2.67</u>	3.71	2.80	0.70	73.78%
	MAPE	<u>0.35</u>	0.83	0.75	0.38	0.68	0.66	0.31	11.43%
CDdidi→BJTaxi	MAE	3.12	3.30	2.00	3.11	3.26	1.82	<u>1.87</u>	-2.67%
	RMSE	4.76	4.91	3.02	4.73	4.87	<u>2.82</u>	2.79	1.06%
	MAPE	0.26	0.62	0.36	<u>0.25</u>	0.56	0.29	0.19	24.00%

5 Experiment

5.1 Experiment Setup

5.1.1 Dataset. We use four widely adopted urban flow datasets for evaluation, BJTaxi, NYCTaxi, NYCBike and CDdidi. The statistics of the datasets are given in Table 1. The details of the datasets are introduced as follows.

- **BJTaxi** contains the taxi trips in Beijing covering from March 1 to June 30 in 2015. We partition the entire data into non-overlapping training, validation and test sets by a ratio of 7 : 2 : 1.
- **NYCTaxi** contains over 160 million taxi trip records in New York from January 1 to December 31 in 2015. Each taxi trip includes pick-up and drop-off dates/times, pick-up and drop-off locations, trip distance, itemized fare, rate type, payment type, and driver reported passenger counts. For this dataset, we use the first 11 months data for training and validation, and the last month data for testing.
- **NYCBike** contains more than 9 million bike trips in New York from January 1 to December 31 in 2015. Each bike trip contains the trip duration, start/end timestamps, and station latitude/longitude. For this dataset, we also use the first 11 months data for training and validation, and the last month data for testing.

- **CDdidi** contains 1 million DiDi taxi trips in Chengdu, China from November 1 to November 31 in 2016. Each trip contains the start-time, end-time, pickup latitude/longitude, and drop-off latitude/longitude. For this dataset, we use 22 days data for training, 6 days data for validation, and the remaining data for testing.

5.1.2 Baselines. We compare FGITrans with two types of baselines. The first type includes the following methods that use a small amount of data from the target city for training its inference model.

- **UrbanPy** [7] is a recent state-of-the-art fine-grained urban flow inference model which employs a pyramid architecture containing multiple components. Each component functions as an atomic upsampler for a small scale, which contains an external factor fusion net, an inference network, a proposal net and a correction net.
- **MT-CSR** [8] conducts urban traffic completion and super-resolution simultaneously under a multi-task learning framework. It employs joint training and end-to-end optimization strategies, and considers the local geographical and global semantic correlations within the data.
- **DiffUFlow** [29] is a diffusion-based fine-grained urban flow inference framework. It designs two unique components to capture spatiotemporal dependencies and urban semantic

features respectively, and integrates the two types of features into the backbone. Each component and backbone are jointly trained in an end-to-end manner in each denoising step.

The second type methods train the above baselines on the source city data and fine-tune them with a small amount of data from the target city. These methods include **UrbanPy(FT)**, **MT-CSR(FT)** and **DiffUFlow(FT)**.

5.1.3 Evaluation Metrics. We evaluate the model performance in terms of Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE), which are defined as follows,

$$MAE = \frac{1}{T} \sum_{t=0}^T |\mathcal{X}^t - \mathcal{Y}^t|, \quad (13)$$

$$RMSE = \sqrt{\frac{1}{T} \sum_{t=0}^T (\|\mathcal{X}^t - \mathcal{Y}^t\|_F^2)}, \quad (14)$$

$$MAPE = \frac{100\%}{T} \sum_{t=1}^T \left| \frac{\mathcal{X}^t - \mathcal{Y}^t}{\mathcal{Y}^t} \right|, \quad (15)$$

where $\mathcal{X}^t$ is the inferred urban flow spatial map at time t and $\mathcal{Y}^t$ is the corresponding ground truth.

5.2 Comparison Result and Analysis

Table 2 presents a comprehensive performance comparison among various methods across four different transfer cases over the four datasets. To ensure a thorough comparison, we evaluate the models by training directly on 20% of the target city’s data as well as by training on all data from the source city followed by fine-tuning with 20% data of the target city. The best results are highlighted in bold, and the second best are underlined for clarity.

The results show that FGITrans achieves the best performance in most cases. In particular, in NYCBike→CDdidi, which has the least data for the target city, the improvement achieved by FGITrans is more significant. Specifically, FGITrans achieves a 17.27% improvement in terms of MAE, a 73.78% improvement in terms of RMSE, and a 11.43% improvement in terms of MAPE, with the MAE and RMSE improvements being the most significant across all four datasets. This indicates that our triple-branch transformer framework can effectively capture inter-city domain invariance and apply it to urban flow inference in the target city.

The result shows that a consistent performance improvement is achieved after transfer learning for all the methods, verifying the effectiveness of transfer learning in improving fine-grained urban flow inference in data-sparse cities. However, the performance improvement of baseline methods is less significant compared with FGITrans, indicating that merely fine-tuning is insufficient to bridge the gap of data distribution shift between two different cities. UrbanFM leverages traffic-related auxiliary information, thus achieving better performance across all cases compared with other baseline methods, which demonstrates that more information does help in urban flow inference.

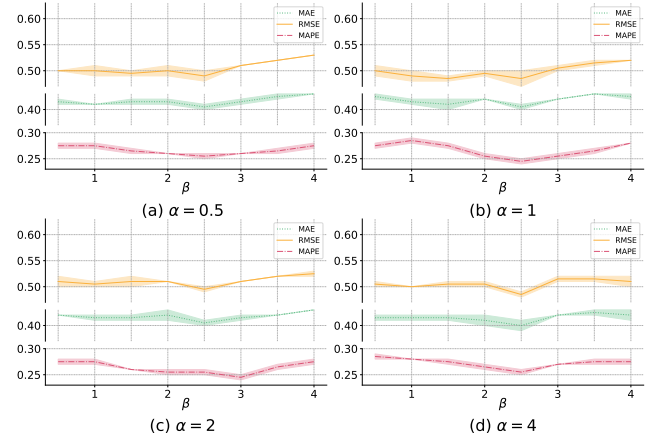


Figure 3: Ablation study on weighting parameters. We repeat each experiment three times and report the average result.

Another important observation is that FGITrans is more robust when transferring knowledge between two dissimilar cities than baselines. Among the four datasets used in our experiment, NYCTaxi and NYCBike are two more similar data because they are from the same city and the covered time periods are both in 2015. This indicates that knowledge transfer between NYCTaxi→NYCBike might be easier, whereas knowledge transfer between other cities might be more challenging. Our results also verify this as UrbanPy and DiffUFlow achieve more significant performance improvement by fine-tuning in NYCTaxi→NYCBike. But in the other three types of transfer methods, the performance improvement after fine-tuning all baseline models is less significant. Specifically, in NYCTaxi→NYCBike, UrbanPy achieves a 30% improvement in terms of MAE, while DiffUFlow achieves a 33% improvement. This indicates that directly transferring the entire city knowledge through fine-tuning is inefficient when the source city and target city are less similar. In comparison, FGITrans consistently achieves lower error than all the baselines, verifying that the knowledge can still be effectively transferred between two dissimilar cities. This is probably mainly because FGITrans can effectively identify inter-city region pairs with similar spatio-temporal patterns (e.g., central business districts), which facilitates cross-city knowledge transfer.

FGITrans shows superior results, suggesting it successfully captures the urban flow patterns between two cities. The performance improvement across all evaluation metrics also indicates that the weight-sharing triple-branch transformer framework and the cross-city knowledge distillation loss are both pivotal in learning the discriminative representations that are both invariant and specific to the domain.

5.3 Ablation Study

We next conduct ablation studies on the case of BJTaxi→NYCTaxi to further evaluate the performance of our model.

Effectiveness of Each Component in CKD loss. As shown in Eq. 12, the CKD loss consists of three parts, the city-aware distillation loss $\mathcal{L}_{CD}$, the pair-wise constraint $\mathcal{L}_{PC}$, and the maximum mean

Table 3: Ablation study on loss components.

Component			Metric MAE / RMSE / MAPE
$\mathcal{L}_{CD}$	$\mathcal{L}_{PC}$	$\mathcal{L}_{MMD}$	
✓	×	×	0.55 / 0.62 / 0.35
✓	✓	×	0.53 / 0.59 / 0.30
✓	✓	✓	0.41 / 0.51 / 0.27

Table 4: Ablation study on different amount of target city data.

	Metrics	UrbanPy(FT)	MT-CSR(FT)	DiffUFlow(FT)	FGITrans
20%	MAE	<u>0.43</u>	1.58	0.97	0.41
	RMSE	<u>0.78</u>	1.95	1.39	0.51
	MAPE	<u>0.29</u>	0.70	0.58	0.27
40%	MAE	<u>0.41</u>	1.52	0.93	0.40
	RMSE	<u>0.76</u>	1.89	1.37	0.49
	MAPE	<u>0.28</u>	0.68	0.55	0.26
60%	MAE	<u>0.40</u>	1.47	0.91	0.36
	RMSE	<u>0.75</u>	1.87	1.34	0.47
	MAPE	<u>0.27</u>	0.67	0.54	0.24

discrepancy loss $\mathcal{L}_{MMD}$, respectively. We study the effectiveness of each component and the results are shown in Table 3.

The result demonstrates the significance of each component, where $\mathcal{L}_{CD}$ helps in narrow the cross-city disparities by transferring resolution-aware knowledge, $\mathcal{L}_{PC}$ enforces a direct feature space alignment between the source and target cities, and $\mathcal{L}_{MMD}$ further refines the alignment by reducing the distributional discrepancies between cities. The result reveals the individual effectiveness of $\mathcal{L}_{MMD}$, which ensures the consistency in feature representation across different cities by minimizing their distributional difference.

Effect of Weighting Parameters α and β . The importance of $\mathcal{L}_{PC}$ and $\mathcal{L}_{MMD}$ is controlled by the weighting parameter α and β . As illustrated in Figure 3, one can observe an overall trend that the performance first increases and then decreases when β increases from 0.25 to 4 while keeping α fixed. However, FGITrans is not sensitive to the changes of α . By referring to the experimental result, we empirically set α to 1 and β to 2.5, which achieves the best result of the experimental metrics.

Effect of Target City Data. We also explore the impact of the amount of data available in the target city on the performance of various models. Besides setting 20% target city data available, we further test the model performance with more target data. Specifically, we make 40% and 60% target city data available. The result of using different amount of target city data is presented in Table 4. It illustrates the consistent superior performance of FGITrans across different amount of available target city data compared to the baselines. As the available data in the target city increases, the performance of all the models improves accordingly. The performance improvement of FGITrans is most significant, which indicates that FGITrans can better leverage its transfer ability when more data is available.

6 Conclusion

In this paper, we proposed a novel cross-city fine-grained urban flow inference model FGITrans to infer the fine-grained urban flow

in data-scarce target cities by transferring knowledge from data-rich source cities. FGITrans addressed two challenges which were significant to cross-city fine-grained urban flow inference, including the insufficient integration of domain-invariant and domain-specific knowledge, and the significant discrepancy between source and target cities. Extensive evaluations on four real large datasets showed FGITrans significantly enhanced the performance and outperformed the state-of-the-art models.

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